

SOFTWARE QUALITY AND CONFIGURATION MANAGEMENT

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SOFTWARE QUALITY MANAGEMENT

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Quality means that a product satisfies the demands of its specifications. It also means achieving a high level of customer satisfaction with the product.

SQM Principles:

- At organizational level: Establishing a framework of organizational processes and standards.
- At project level : have a check on planned process and matching output with standards.
- Quality management at the project level is also concerned with establishing a quality plan for a project

SQM Activities:

- Quality assurance and standards
- Quality planning
- Quality control

QUALITY PLANNING

- A quality plan sets out the desired product qualities and how these are assessed and define the most significant quality attributes.
- Structure of quality management:
 - Product introduction
 - Product plans
 - Process descriptions
 - Quality goals
 - Risks and risk management
- Quality management attributes:
 - 1.Safety 2.Understandability 3.Security
 - 4.Reliability 5.Adaptability 6.Portability
 - 7.Accessibility 8.Reusability etc..

QUALITY CONTROL

- Checking the software development process (within a particular project) to ensure that procedures and standards, as defined in the quality plan, are being followed
- Two approaches to quality control
 - (Manual) Quality reviews – main approach
 - (Automated) Quality measurement
- Types of reviews:
 - Design or program inspection. (part of software validation)
 - Progress reviews. (part of project)
 - Quality reviews. (part of quality management)

SOFTWARE CONFIGURATION MANAGEMENT

Configuration management (CM) is a systems engineering process for establishing and maintaining consistency of a product's performance, functional, and physical attributes with its requirements, design, and operational information.

What is the need of configuration management??

SCM activities:

- Configuration management planning
- Change management
- Version and release management
- System building

CM Standards

CM Planning

- Starts during the early phases of the project
- All products of the software process may have to be managed
 - Specifications
 - Designs
 - Programs
 - Test data
 - User manuals

CHANGE MANAGEMENT

- Software systems are subject to continual change requests:
 - From users
 - From developers
 - From market forces
- Change management is concerned with keeping track of these changes and ensuring that they are implemented in the most cost-effective way.
- Deciding factors for approval of change are:
 - The consequences of not making the change
 - The benefits of the change
 - The number of users affected by the change
 - The costs of making the change

VERSION AND RELEASE MANAGEMENT

Version management

- Version management (VM) is the process of keeping track of different versions of software components or configuration items and the systems in which these components are used.
- It also involves ensuring that changes made by different developers to these versions do not interfere with each other.
- Feature of version management:
 - Version and release identification
 - Storage management
 - Change history recording
 - Independent development
 - Project support

RELEASE MANAGEMENT

- Releases incorporate changes forced on the system by errors discovered by users and by hardware changes
- They also incorporate new system functionality
- Release planning is concerned with when to issue a system version as a release
- System release's : what it is and what does it includes.
- Release problems.
- Release creation